C PROGRAMMING – MODULE-WISE QUESTION BANK

IAT-01

Modules 1, 2 and 3

MODULE 1: Introduction to Problem Solving, Computational Thinking and Fundamentals of C Programming

1. Define computational thinking and state its core pillars used in problem solving
2. What is a flowchart? Explain the commonly used flowchart symbols and their purposes.
3. What is an algorithm? State any two essential characteristics that every valid algorithm must possess.
4. Explain the overview and important features of the C programming language.
5. Explain the basic structure of a C program with a suitable example.
6. Explain the role of a compiler in the C program execution process.
7. Differentiate between keywords and user-defined identifiers in C with one example of each.
8. What are variables and constants in C? Explain their declaration and initialization with examples.
9. Explain the different data types available in C with suitable examples.
10. Explain arithmetic, relational, logical and assignment operators in C with examples.

MODULE 2: Decision Making and Iterative Problem Solving

11. Explain the if-else statement with syntax and example.
12. Explain the working of for, while, and do-while loops in C with their respective syntax structures. Also, compare the primary difference between entry-controlled and exit-controlled loops using a suitable example.
13. Write a C program to check whether a given number is positive, negative or zero.
14. Write a C program that accepts an integer choice and two integer numbers from the user, then uses a switch-case statement to perform addition, subtraction, multiplication, or division based on the choice, handling division by zero appropriately.
15. Write a C program to print a square and a right-angled star pattern using nested loops.
16. Write a C program to print a pyramid pattern using nested loops.

MODULE 3: Arrays and Strings

17. What is an array? Explain the need for arrays in C programming.
18. Explain the difference between one-dimensional and two-dimensional arrays in C with respect to their declaration, memory representation, and initialization.
19. Write a C program to accept a two-dimensional integer matrix of order M×N, display the entered matrix, and compute the sum of all elements present on its main diagonal.
20. What is a string in C? Explain string declaration and initialization with examples.
21. Explain any three built-in string handling functions available in the <string.h> header file, providing the exact syntax and purpose of each function.
22. Write a C program that uses string manipulation functions from string.h to read two strings, concatenate them, and display the length of the resulting string.